Yan Wang

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EDUCATION

Wuhan University, Wuhan, China *Sep. 2023-Present*

B.S. in Biological Science; **GPA: 3.91/4.00; Rank: 1/33; Major GPA: 3.97/4.00**

Selected coursework: Cell Biology(95) Molecular Biology(96) Genetics(96) Biochemistry(95)

TOEFL iBT: 106/120 (Reading 29, Listening 28, Speaking 24, Writing 25)

Yale University, New Haven, CT, USA *Expected Aug. 2026-May. 2027*

Visiting Undergraduate Researcher, Scherzer Lab, Adams Center for Parkinson's Disease Research

Tsinghua University High School, Beijing, China *Sep. 2020-Jun. 2023*

National College Entrance Examination, **Biology:100/100**

RESEARCH INTERESTS

1.Neural coding underlying higher cognitive functions like learning, memory, social, decision, etc.

2.Hypothalamic regulation of reversible hypometabolism, thermoregulation, arousal, and systemic energy conservation during induced torpor/hibernation.

RESEARCH EXPERIENCE

Medical Research Institute, Wuhan University *Jan. 2025-Present*

Joint lab of Prof. Yanxun Yu and Prof. Youngnam Jin

Wang, Y.*, Zhang, J., Huang, X., & Yu, Y. Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER-AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*. *Neuroscience 2026, Society for Neuroscience*. Scientific Abstract.

- Led a *C. elegans* neurobiology project investigating how insulin-like neuropeptide(INS-1) signaling assigns negative valence to ethanol under high-salt conditions.
- Discovered that *ins-1* loss completely reverses high-salt ethanol avoidance into attraction, identifying INS-1 as a key neuromodulator of context-dependent behavioral valence.
- Mapped distributed INS-1-releasing nodes by combining cell-specific *ins-1* deletion, rescue, and neuronal ablation, implicating ASEL, AIY, AIZ, and AIB in ethanol avoidance.
- Resolved ASER-AIY circuit dysfunction using calcium imaging, showing a right-shifted ASER ethanol response range and weakened AIY negative calcium transients in *ins-1* mutants.
- Demonstrated that ASEL-specific *ins-1* rescue significantly shifts the ASER ethanol response range back toward wild type, supporting ASEL-derived INS-1 as a major regulator of ASER sensory tuning.
- Integrated ethanol dose-dependent behavioral assays with physical modeling of ethanol diffusion and sensory-threshold shifts, supporting a quantitative model in which INS-1 tunes ethanol sensitivity rather than causing simple sensory loss.
- Proposed NTR-1 as a candidate noncanonical INS-1 receptor by integrating pathway mutant analysis, *ntr-1* phenotyping, and molecular docking.
- Developed a circuit-level model in which distributed INS-1 release, potentially mediated by NTR-1, calibrates ASER sensory gain and downstream AIY output to code ethanol valence under high-salts.

Project title: Performance Optimization of the POST-IT Proximity Tagging System

- Second researchers optimizing POST-IT, a Pupylation-based non-diffusive proximity-labeling platform for live-cell-compatible small-molecule target identification.
- Engineered PafA ligase to reduce the major kinetic and substrate constraints of the original POST-IT system, including long labeling time and high SBP–Pup(E) requirement.
- Built a yeast-surface-displayed Pep1-MATH weak-interaction selection system to couple PafA catalytic activity with FACS-based enrichment of improved variants.

- Generated a ~10^6-scale PafA mutant library by error-prone PCR and performed iterative directed evolution under increasingly stringent labeling conditions.
- Combined FACS screening, NGS-guided mutation analysis, and combinatorial reconstruction to obtain high-performing multi-mutant PafA variants.
- Improved labeling from 18 h / 40 μM to 2 h / 10 μM, increasing threshold-positive cells from 0.5% to 27.2% and achieving >40-fold apparent performance enhancement.
- Conducted molecular cloning, yeast transformation, protein expression/purification, SDS-PAGE validation, flow cytometry, confocal imaging, and AlphaFold/PyMOL-based structural analysis.

College of Life Science, Wuhan University *Nov. 2024-Oct. 2025*

Project title: LOGIC -Lasting Optimized Genetic Input Calculator, 2025 IGEM Gold Medal

- Original proposer and core system designer of LOGIC, a Gold Medal-winning IGEM project that reconceptualized quorum-sensing-mediated intercellular communication as programmable spatial "wiring" for colony-level biological computation.
- Designed a diffusion-based biocomputing architecture in which orthogonal AHL signaling channels, engineered sender-receiver colonies, and spatial concentration gradients were integrated to implement modular genetic logic beyond single-cell circuit constraints.
- Built and characterized high-pass and band-pass response modules to encode Boolean operations, constructed AND, OR, XOR, multi-input gates, and cascaded logic operations on solid media.
- Developed a serial biological adder framework by coordinating input, output, and carry signals, advancing the system from isolated logic gates toward executable biological arithmetic.
- Integrated optogenetic refresh modules, including split-VVD AiiA for light-controlled AHL degradation and LOVdeg-tagged output proteins for signal reset, together with modeling and >20,000 fluorescence readouts to assemble a reusable LOGIC toolbox.

TECHNICAL SKILLS

- Microbial and model organism handling: *E. coli*, yeast, and *C. elegans*.
- Molecular cloning, protein purification, Western blot, CO-IP, RT-qPCR, FACS, and routine biochemical assays, mammalian cell culture, etc.
- Microinjection, crosses, genetic manipulation, calcium imaging, optogenetic manipulation, and behavioral assays in *C. elegans*.
- C, HTML, CSS, etc. Familiar with basic single-cell RNA-seq data analysis workflows in R.
- Basic structural biology tools (AlphaFold and PyMOL), data analysis, data visualization.

ACADEMIC ACTIVITIES

Tsinghua University, Beijing, China *May. 2025*

National Forum on China's "Top-Tier Talent Program 2.0" in Biology; one of two **WHU delegates**.

Westlake University, Hangzhou, China *Jul. 2025-Aug. 2025*

"Frontiers of Life Sciences" International Undergraduate Summer School

Peking University / Center for Life Sciences, Beijing, China *Jul. 2026*

2026 Summer Training Program in Neuroscience and Cognitive Science

HONORS & AWARDS

Gold Medal, International Genetically Engineered Machine Competition (iGEM); *Nov. 2025*

Gold Prize , Regular Track, SynBio Challenges (an international contes) *Aug. 2025*

Liu Daoyu Creative Learning Scholarship (awarded to 24 students university-wide) *Jan. 2026*

Hongyi Honors College's Annual Top Ten Science Advances Award *Oct. 2025*

First-Class Outstanding Freshman Scholarship (2023); Outstanding Student Scholarship (5%, 2024 & 2025); Outstanding Student (5%, 2024 & 2025); Outstanding Student Leader (2024 & 2025*)*